



UNIVERSITY OF AMSTERDAM

Social Media Analytics: Natural Language Processing & Text Mining

Current Advances: PARA-Language & Emoji Analytics

Generative AI & Large Language Models

Digital Marketing and Analytics: Week 3

Academic Year 2025-2026 – Semester 1 – Block 2

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Business

Text Mining & Analytics:

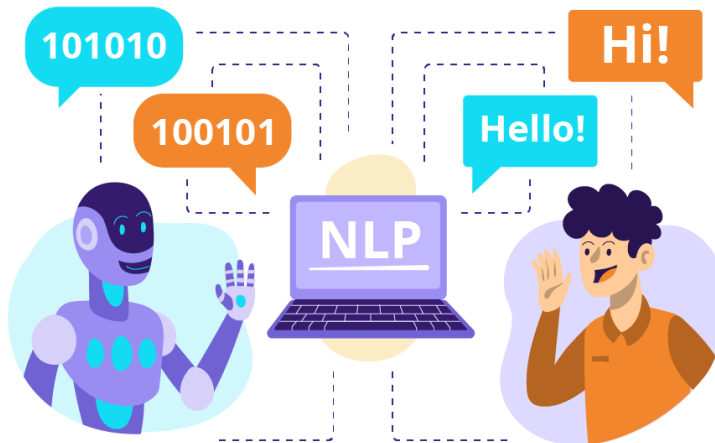
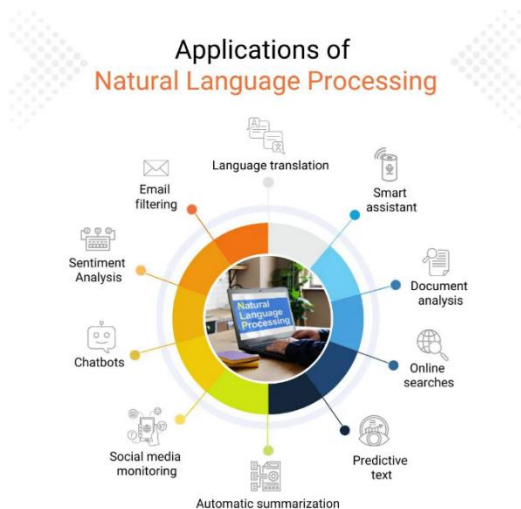
NLP: Sentiment and Topic Analysis

Natural Processing Language (NLP)

NLP is a branch of artificial intelligence that focuses on the interaction between computers and human languages.

NLP uses computational techniques to **analyze, understand, and generate human language**.

It helps computers understand the natural language input and generate natural language output.



NLP & Textual Data Analysis: Sentiment Analysis

Sentiment Analysis: the process of computationally identifying and categorizing opinions expressed in text, to determine whether the attitude towards a particular topic, product, etc. is positive, negative, or neutral.

In sentiment analysis, social conversations are classified based on **positive**, **negative** and **neutral** language.

However, automated sentiment analysis will never be 100% accurate due to the many subtleties of human expression



She THANKS to the firm for kicking out a family – and states that she wants to fly more with the firm
But is this really a POSITIVE comment ?



NLP & Textual Data > Sentiment Analysis – How it Works ?

Now you can have a numeric transformation for this text: 1 – positive or 0 – negative : in your data

This will be coded as one positive statement (1) since the statement (as a whole) marked as **positive** in sentiment analysis.

Consider this problem instance: "Sam is a great guy."



- Tokenize:



NLP & Textual Data > Sentiment Analysis – How it Works ?

To have score-based sentiment scores rather than merely coding 0-1 (negative-positive):

Each word is scored on the basis of its sentiment based on a **Lexicon database**: this is **Lexicon Based approach (tutorial 2)**

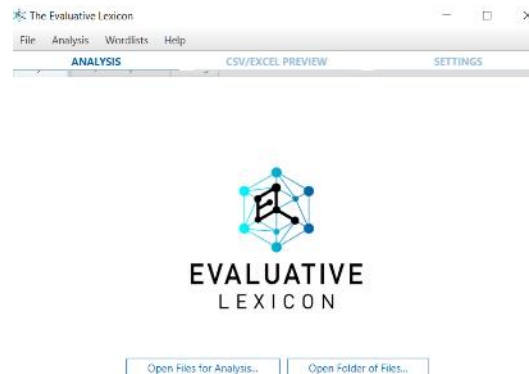
每个词都根据其情感倾向，并基于词典数据库进行评分：这就是基于词典的方法（教程 2）。

Then you would probably have comments/tweets scored as:

+9,2 > very positive
+1,7 > slightly positive
-1,4 > slightly negative

Word	Sentiment	Valence/Intensity
okay	Positive	0.9
good	Positive	1.9
great	Positive	3.1
horrible	Negative	-2.5
:(	Negative	-2.2
Sucks (or the slang: sux)	Negative	-1.5

*There are many Lexicon based tools/software
“Evaluative Lexicon” is a good usable example.*



need to download!

NLP & Textual Data > Sentiment Analysis – How it Works ?

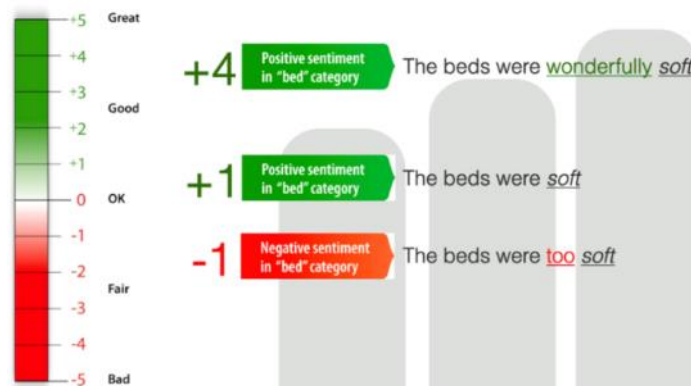
See the example to have an idea about possible complications in sentiment analysis:

IF the comment reads as “the beds were too soft” – we know it is a negative comment

IF the comment reads as “the beds were wonderfully soft” – we know it is a positive comment

But what if the comment reads only: “the beds were soft” ? > Do you think it is “neutral” ?

F	G	H	I	J
location	sentiment	contents	authname	
Omaha, Neb	-1	Starbucks computer glitch means free drinks http://t.co/cD4lf6GHaj	KETV NewsWatch 7	
	0	RT @ArianaGrande so starbucks is closing i'm pregnant my nudes leaked & i see a...		
Stark County	0	RT @sanctuarymg Will you be cheering at the @YMCAStark #NorthCanton	YMCA of Cntrl Stark	
valdivia - chil	0	DHhh muy cansada pero alfin en starbucks ðŸ™ˆ	frafrafran	
Tampa, FL	1	I live off of Starbucks Arizona tea and blue cherry Gatorade ðŸ™ˆ...	Sky	
	1	RT @camilacabello97 today i walked around New York City with a hot Starbu	LERN JERGI	
Hawthorne, e	0	@JoslynnLoren Starbucks on el segundo and hawthorne	senpai	
	1	Time to relax ðŸ™ˆ #Starbucks http://t.co/u2oEkloMdv	Elizabethâ	
Washington	1	@Alex_is_coded within a year you could be a manager if you work hard enou	Versace Princess	
Cosby, TN	0	RT @EverythingGoats I goat some starbucks â™ˆ•ï¸¸ ðŸ™ˆ http://t.co/6u2J	Jordan Self	
	1	RT @Luluss_v76 I want a vanilla bean frappe from Starbucks	â™ˆ T, â™ˆ T, â™ˆ T, â™ˆ T	
	0	RT @AminePosey Vous allez au Starbucks pour la boisson ou pour Instagram	EL MAESTROâ€œ	
Buenos Aires	0	“Yo que querâ€a ir a starbucks con con vosccc” que linda que es como la quâ™ˆ†		
Hawaii Pacifi	0	I'm the asshole that asks Coffee Bean if they have (a variation of) a Starbucks	Michelle C	
	1	I'm so thankful I live so close to a Starbucks ðŸ™ˆCE	Diana.	
	-1	No matcha at this Starbucks ðŸ™ˆa	Melissa S.	
North Missis	-1	Never been to a Starbucks. Who the hell wants high priced coffee?	Cox Da Gamer	
		RT @LetranCollege #LetranFroshTips kung walang perang pang Starbucks me Ra		
		RT @InaAraujo99 Apetece-me starbucks	clAjudia sofia	
		@sillysampi starbucks itu pake bubuk yg dilarutin.	the wicked bitch	
		RT @Starbucks Proud. #LoveWins http://t.co/ADDXKlyhcb	Angelina	
Hermosillo Sonora		Starbucks repara falla tÃ©cnica y reanuda el servicio http://t.co/Z5WwgyfzV	Rebecca Dessens	
italy		@Ciccio8a ti farÃ© la musica che sparano qui da Starbucks. che te ne fai del Qui The new londoner		



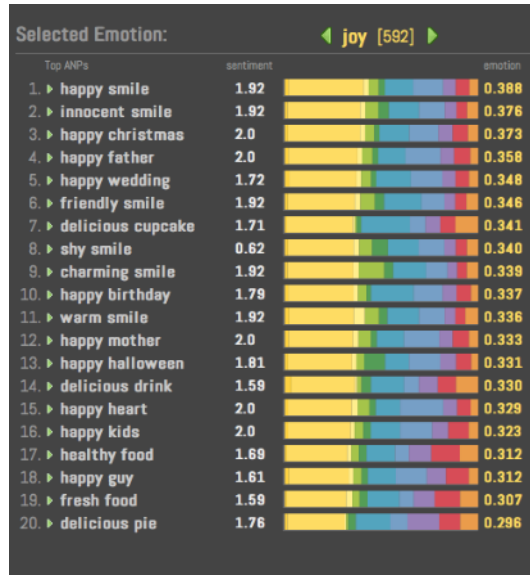
Now let's think again – *why would someone make a voluntary statement about beds* – when her/his *view on beds* are really neutral ?

Beyond Sentiment: Towards Semantics of Emotions

Positive-Negative dichotomy falls short in capturing human emotions.

What EMOTIONS are expressed by people/customers ?

Thus: A Semantic analysis would give a better picture on what people actually think and therefore would be more predictive.



Textual Data: Multi-Layered Sentiment Analysis

除了仅仅关注积极性之外，基于机器学习的工具还能帮助从情感的不同维度对文本数据进行评分。

Rather than merely looking at positivity ML based tools also help in scoring the textual data on different dimensions regarding sentiment

- 1 *Valence* *positivity-negativity*
- 2 *Subjectivity (Emotionally)* *to what extent the text is factual or subjective: emotion-opinion based*
- 3 *Polarity (Extremity)* *how strong the positivity in (1) is expressed*

Sentiment Analysis	Subjectivity Analysis
Positive	Subjective
Negative	
Neutral	Objective

Table 3: Text examples and their complexity scores from a Tumblr blog.

Text body	Topic-complexity	Sentence-complexity
You've been slicing, dicing, pitting and peeling your produce all wrong. Until now. Cooking is fun. Eating is fun. Sometimes you want to get the cooking done so you can get to the eating. We can respect that. See all of our cooking hacks that will save you time in the kitchen here.	0.558	0.437
Instant close-up shots of people and objects courtesy of the Lomo'Instant. We know you're already raring to get your hands on the Lomo'Instant, and to help you tide over until it arrives at your doorsteps, we have these test shots to show you straight from the Lomography team in Hong Kong!	1.583	0.820

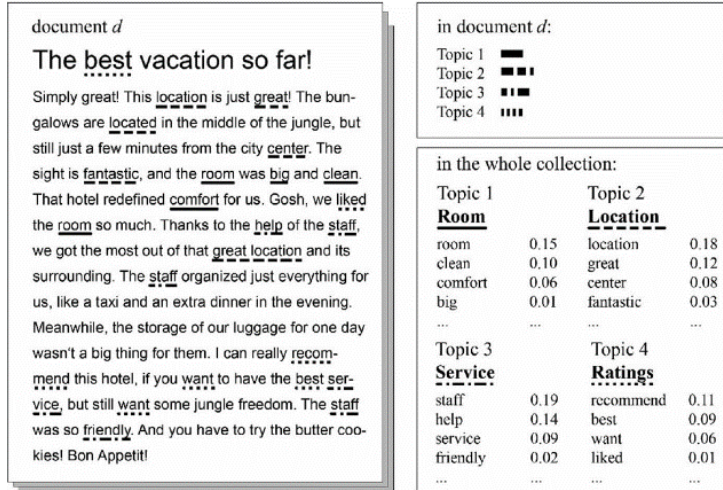
Textual Data > Topic Analysis – What Topics are There ?

主题模型提取并分类（聚类）文本数据（评论、社交媒体帖子等）中提及的主要主题。

它们还用于对社交媒体、客户服务热线和在线评论中的关键主题进行细分和聚类。

Topic Models extracts and classifies (clusters) the **prominent TOPICS** mentioned in a textual data (reviews, social media posts etc)

They are also used to segment-cluster the essential topics in Social Media, Customer Service Lines and Online Reviews.



Topics found by using 1,8 million articles from NY Times by using Topic Modeling/Clustering (LDA)

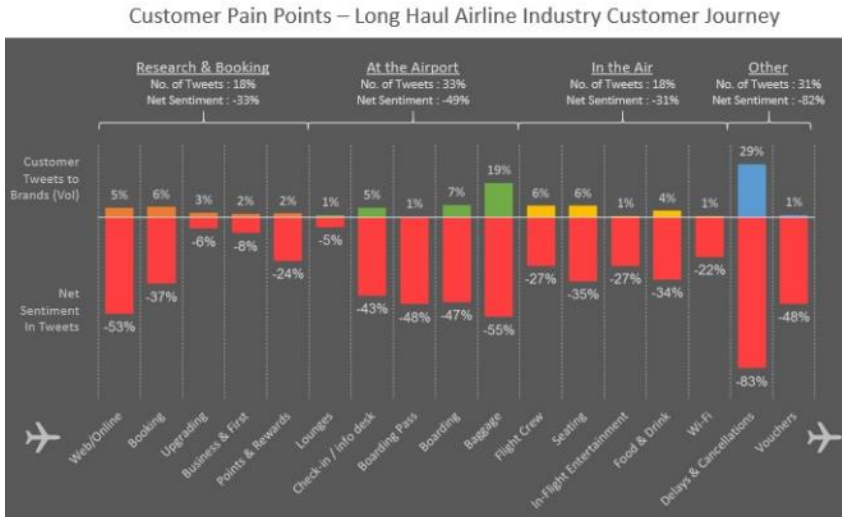


Example 1: Sentiment & Topic Analysis Used Together

Customer Journey Maps

Topic and Sentiment analysis: used to manage the pleasure-pain points along **CUSTOMER JOURNEYS**

Check the example: The rate (%) of **positive-negative comments** in **each stage** of customer journey based on **each TOPIC**



Source: Twitter n=25,097 Period: 1st Jan to 10th Jan 2017

Brands included: Air Canada, Air New Zealand, American Air, BA, Cathay Pacific, Emirates, Etihad, Qantas, Qatar Airways, Singapore Air, United, Virgin Atlantic

AIRLINES CONCERNS

about United Airlines on Mar 14, 2018



Negative sentiment for United is still extremely high today at 52 percent.

SENTIMENT

about United Airlines on Mar 14, 2018

DEEP COMPARISON OF SENTIMENTS >



Example 2: NLP-Processed Text Data as an Input in Model

What Holds Attention. Linguistic Drivers of Engagement

What makes some content more engaging? A multimethod investigation combines controlled experiments with Natural Language Processing of 600,000 reading sessions from over 35,000 pieces of content to examine what types of language hold attention and why

*Findings: **Anxious, Exciting, Anger and Hopeful** language holds attention while **Sad** language discourages it.*

***Concreteness, Familiarity, Readability and Simplicity** also drives Customer Engagement.*

	Emotionality (1)	Valence (2)	Specific Emotions (3)
Emotion			
Emotionality	.007*** .001		
Positive emotion		.023*** .002	.025*** .002
Negative emotion		.006*** .001	
Anger			.011*** .001
Anxiety			.042*** .001
Sadness			-.036*** .001
Processing Ease			
Readability	.065*** .001	.066*** .001	.070*** .001
Syntactic simplicity	.051*** .001	.05*** .001	.051*** .001
Familiarity	.028*** .001	.028*** .001	.029*** .001
Concreteness	.048***	.051***	.046***

Most liked instagram post by L. Messi

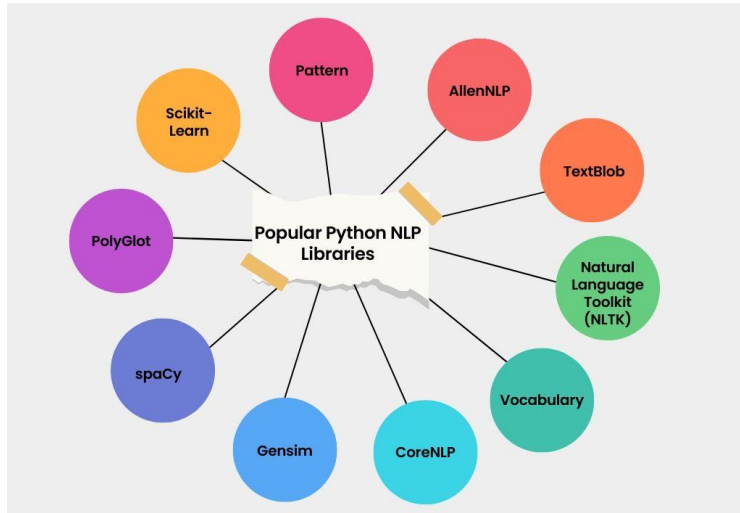


Natural Processing Language (NLP) – Practical Start with NLTK-Spacy

The Natural Language Toolkit (NLTK) is a Python package for natural language processing.

Last week, in Tutorial 2 – Part 2 you already did some practical text-analytics by using Text Blob and Hugging-Face.

NOW Let's practice with NLTK to learn the basics/fundamentals of text analytics to get more into the simple steps and logic in text analytics.

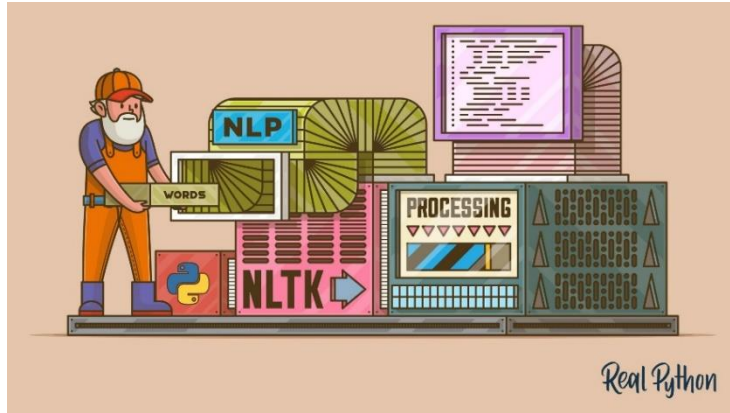


Jupyter Notebook is a web application that allows you to combine computer code, formatted text , and visualizations in a single document.

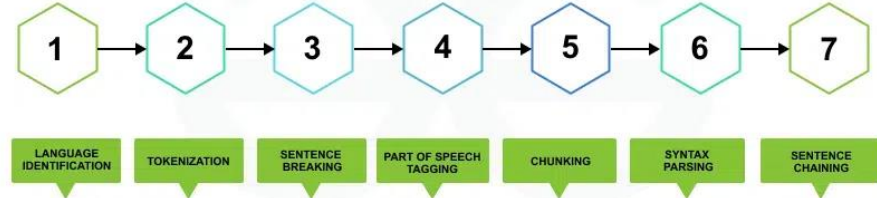
Natural Processing Language (NLP) – Practical Start with NLTK-Spacy

NLTK is a well-known and multi-purpose package for Text Analytics. Advantage: Support many languages other than English.

- [CLICK HERE](#)
- Create a copy of the Jupyter notebook
- Edit -> Select all cells -> Copy cell or selection > File -> New notebook -> Paste
Read the text and start running the Python code following the sequence.



7 Step Text Analytics Process



Textual Paralanguage:

Applications & Use in Digital Marketing

Para-Linguistics: Non-Verbal Cues in Text Data – Emojis and Beyond

副语言：指言语中非实际口头表达的部分，它提供语境信息，使互动者能够更恰当地理解所传达的信息。

Paralanguage: Aspects of speech that are not actual verbal prose, gives contextual information that allows interactors to more appropriately understand the message being conveyed.

1 Auditory Non-Verbal Communication:

information communicated by aspects of speech such as pitch, rhythm, tempo, vocal qualities and vocalizations

2 Visual Non-Verbal Communication:

conscious or unconscious bodily movements that possess communicative value, including human gestures and body language.

3 Tactile Non-Verbal Communication:

nonverbal communication related to physical, haptic interaction with another individual



注意: VADER_FOR emojis_pyhon software

Article 2.3: Berger, J. et al. (2022). Wisdom from words: marketing insights from text. Marketing Letters, 1-13.

Luangrath, A. W. et al. (2017). Textual paralanguage and its implications for marketing communications. Journal of Consumer Psychology, 27(1), 98-107.

Para-Linguistics: Non-Verbal Cues in Text Data – Emojis and Beyond

Emoji Marketing:

Emojis, or pictographs that supplement or replace written language, have become ubiquitous in contemporary communication.

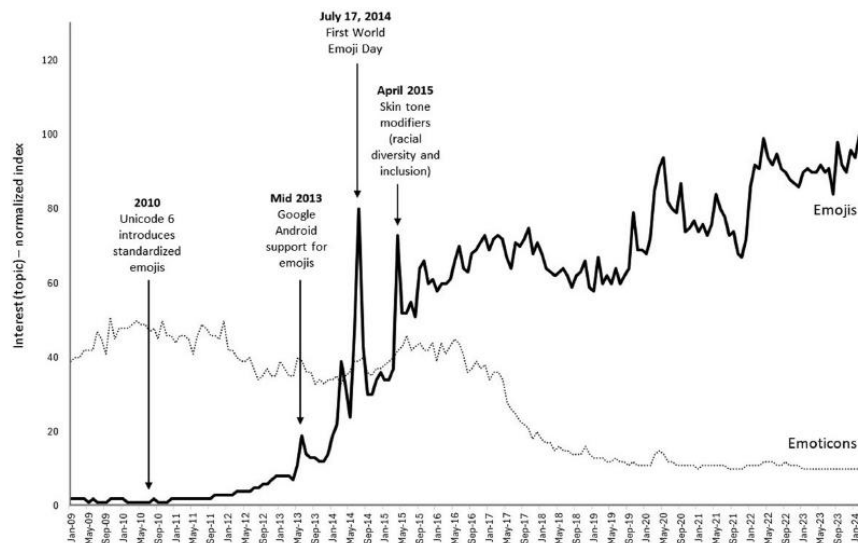


Fig. 1. Shifting importance of emoticons and emojis based on Google Trends analysis (worldwide).



Iconic vs Symbolic Use of Emojis

Iconic Emoji: directly represents the construct of interest by exhibiting the same structure,

Slice of pizza emoji in a message promoting a discount from a pizza restaurant would represent an iconic emoji use;



Symbolic Emoji: represents a higher-order, abstract construct that requires interpretation and elaboration beyond the image

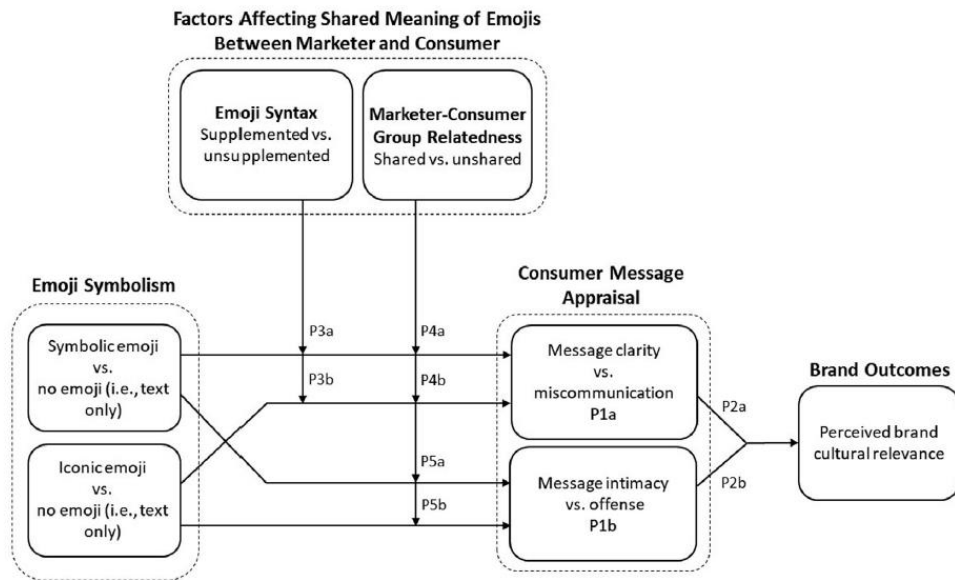
Tree emoji in a promotional message for electric cars likely aims to symbolize environmental protection or sustainable consumption behaviors, which requires consumer interpretation.



Emoji Marketing – Conceptual Framework

Emoji Syntax: The use of emojis to supplement text or in a stand-alone (substitute) mode without accompanying text.

Marketing-Emoji Group Relatedness: The extent to which communication is restricted to ingroups (shared group membership) or directed toward a more general, non-targeted audience (unshared group membership).



Emoji Syntax: Supplemented Complements with Text

Sadly, no meal on this flight 😞

Emoji Syntax: Stand Alone (Unsupplemented)

😂 + 🎄 + 📱 + ☕ + 😊

Group Relatedness:

the emoji sequence “👤🐶📝💰”

Fortnite online game users – not known to general audience

Emojis & Emotion Analytics

- The three modalities (text, emoji and images) encode different information to express emotion and therefore can complement each other.
- Emoji sense depends on the textual context, and emoji combined with text encodes better information than considered separately.

Emotion \ Emojis	1	2	3	4	5
Anger	🖤	🔥	💥	😡	💔
Fear	👻	💩	💀	👹	😱
Joy	❤️	😄	😁	💜	🎵
Love	❤️	😍	💕	😘	💙
Sad	💔	😞	😓	😭	🥀
Surprise	😲	😄	💖	😁	💫
Thankful	❤️	💖	🙏	😁	💖



Source: Illendula, A., & Sheth, A. (2019, May). Multimodal emotion classification. In *companion proceedings of the 2019 world wide web conference* (pp. 439-449).

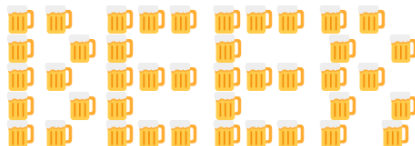
Emojis & Research

- 在数字市场中，非面部表情符号 (NF) 比面部表情符号更能有效地提升口碑传播量。
- 使用多个替代的非面部表情符号虽然可以节省字符，但最终会降低信息处理流畅度（边界条件）。
- 对于优质卖家而言，使用多个非面部表情符号可能会降低口碑传播量，因为这会降低消费者对其能力的感知。

- Non-face (NF) emojis can be more effective than face emojis in driving eWOM volume in digital marketplaces
- Using multiple substitutive NF emojis may save characters but ultimately reduces processing fluency (boundary condition)
- For premium sellers, using multiple NF emojis may reduce eWOM volume due to reduced perceptions of competence



We'll bring the



NFL @NFL · May 31



days until #Kickoff2022!

10:35 PM · May 31, 2022 · Twitter for iPhone

23 Retweets 2 Quote Tweets 163 Likes



Not Wordle, just a Double Stuf OREO

1:57 AM · Jan 27, 2022 · Twitter for iPhone

831 Retweets 88 Quote Tweets 8,367 Likes

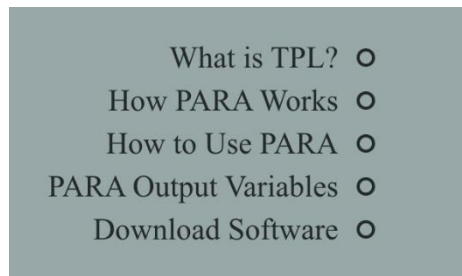


Source: Orazi, D. C., Ranjan, B., & Cheng, Y. (2023). Non-face emojis in digital marketing: Effects, contingencies, and strategic recommendations. *Journal of the Academy of Marketing Science*, 1-28.

Paralanguage & Emoji Analytics in Practice: PARA software

PARA is a free scientifically-validated software to **analyze non-verbal cues (such as emojis) in textual language**. Very useful in social media analytics.

PARA Software: [Click here](#)



Step 1:
Click "Download Software"

Download Software

First Name * Last Name *

Email *

We'll never share your information with anyone else.
University/Organization Name *

Organization Type *

Please select your operating system *

Step 2: Fill in the form

After downloading it, double click to open it

Upload File

☐ First row is header

Status: Waiting for file upload.

Step 4: Upload your csv file

	id	author_id	created_at	text	multimodal	lang	like	
1	164802223...	58561993	2023-04-17...	like if you ...	1	en	102	12
2	1647967011...	58561993	2023-04-17...	There are no...	2	en	40	4
3	1646927799...	58561993	2023-04-14...	don't worry...	1	en	47	6
4	1646876467...	58561993	2023-04-14...	https://t.co/...	4	zxx	151	13
5	164669359...	58561993	2023-04-13...	Deihttps://...	1	en	30	2
6	164654276...	58561993	2023-04-13...	you've been ...	1	en	57	12
7	1646195979...	58561993	2023-04-12...	This is your ...	1	en	28	3
8	1646167038...	58561993	2023-04-12...	shhh! he's ...	1	en	101	15
9	1645804017...	58561993	2023-04-11...	drop your pe...	1	en	43	17
10	1645523267...	58561993	2023-04-10...	it's ...	3	en	71	6

Upload File Analyze Save Results

☒ First row is header

Status: Select a column and click analyze.

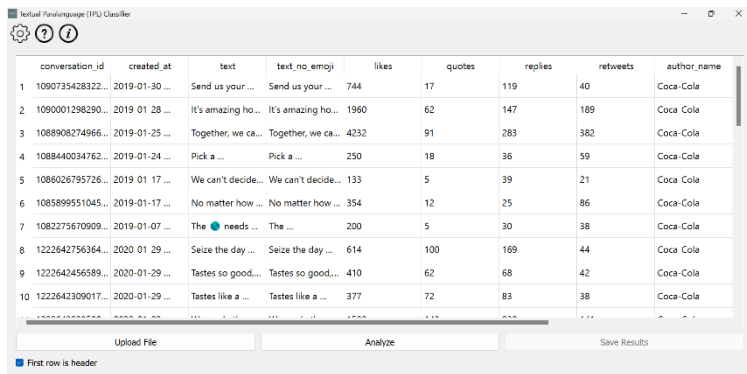
Step 5: Select "First row is head"
AND select the column you want
to analyse the presence of
Paralanguage. In the example, we
Selected the column "text"

Paralanguage & Emoji Analytics in Practice: PARA software

Processing the data

Step 6: Click “Analyze”

Make sure the column with emojis are selected.



conversation_id	created_at	text	text_no_emoji	likes	quotes	replies	retweets	author_name
1090735428322...	2019-01-30 ...	Send us your ...	Send us your ...	744	17	119	40	Coca-Cola
1090001296290...	2019-01-28 ...	It's amazing ho...	It's amazing ho...	1960	62	147	189	Coca-Cola
1088908274966...	2019-01-25 ...	Together, we ca...	Together, we ca...	4232	91	283	382	Coca-Cola
1088440034762...	2019-01-24 ...	Pick a ...	Pick a ...	250	18	36	59	Coca-Cola
1086026795726...	2019-01-17 ...	We can't decide...	We can't decide...	133	5	39	21	Coca-Cola
1085899551045...	2019-01-17 ...	No matter how ...	No matter how ...	354	12	25	86	Coca-Cola
1082275670909...	2019-01-07 ...	The 🟢 needs ...	The ...	200	5	30	38	Coca-Cola
1222642756364...	2020-01-29 ...	Seize the day ...	Seize the day ...	614	100	169	44	Coca-Cola
1222642456589...	2020-01-29 ...	Tastes so good...	Tastes so good...	410	62	68	42	Coca-Cola
1222642309017...	2020-01-29 ...	Tastes like a ...	Tastes like a ...	377	72	83	38	Coca-Cola

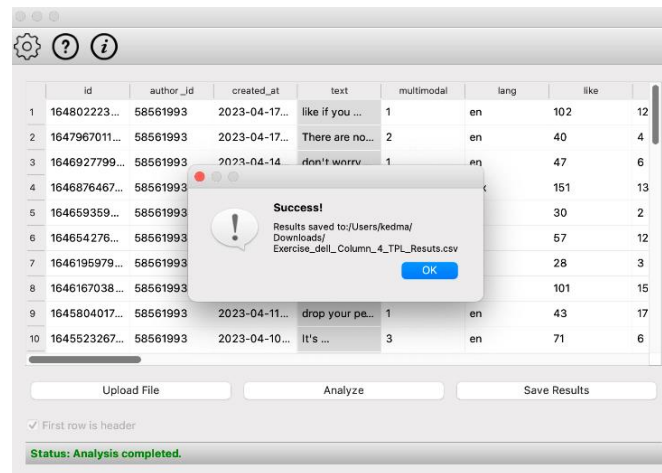
Upload File Analyze Save Results

First row is header

Step 7: The software will tell you when the analysis is completed.

Then, click “Save Results”. PARA will save the results adding TPL in the name.

The results file should now be in your computer. **Well done!**



id	author_id	created_at	text	multimodal	lang	like	like
164802223...	58561993	2023-04-17...	like if you ...	1	en	102	12
1647967011...	58561993	2023-04-17...	There are no...	2	en	40	4
1646927799...	58561993	2023-04-14...	don't worry	1	en	47	6
1646876467...	58561993					151	13
164659359...	58561993					30	2
164654276...	58561993					67	12
1646195979...	58561993					28	3
1646167038...	58561993					101	15
1645804017...	58561993	2023-04-11...	drop your pe...	1	en	43	17
1645523267...	58561993	2023-04-10...	It's ...	3	en	71	6

Upload File Analyze Save Results

First row is header

Status: Analysis completed.

Paralanguage & Emoji Analytics in Practice: PARA software

If you are not sure what do specific terms in your outcome-data means > Take a look at PARA software or website for detailed information:

Superordinate Level Sensory Features	Auditory		Tactile	Visual		Aggregate Variables
Textual Paralinguistic Categories	Voice Qualities	Vocalizations	Tactile Kinesics	Visual Kinesics	Artifacts	
Subordinate Level Nonverbal Features	<u>Pitch</u> <u>Rhythm</u> <u>Stress</u> <u>Emphasis</u> <u>Tempo</u> <u>Volume</u> <u>Censorship</u> <u>Spelling</u>	<u>Alternants</u> <u>Differentiators</u>	<u>Tactile Emojis</u> <u>Tactile Emoticons</u> <u>Alphahaptics</u>	<u>Bodily Emojis</u> <u>Bodily Emoticons</u> <u>Alphakinesics</u>	<u>Nonbodily Emojis</u> <u>Nonbodily Emoticons</u> <u>Formatting</u>	<u>Emoji_Count</u> <u>Emoji_Index</u> <u>Emoticon_Index</u> <u>TPL_Index</u>

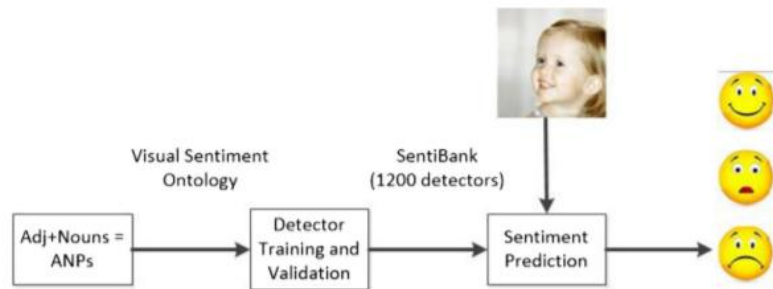
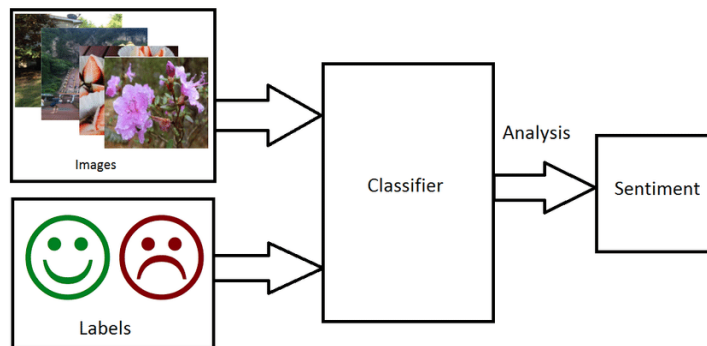
Voice Qualities	Characteristics of the sound of the words being communicated that have to do with how the words should be spoken. Example: This coffee is amazingggggggg
Vocalizations	Utterances, fillers, terms, or sounds that can be spoken or produced by the body that result in an audible noise that is recognizable (not necessarily a “word”) Example: This coffee is amazing. Mmmm
Tactile Kinesics	Nonverbal communication related to physical, haptic interaction with others. Example: This coffee is amazing. 🤝
Visual Kinesics	Nonverbal communication related to movement of any part of the body or the body as a whole. Example: This coffee is amazing. 🙌
Artifacts	The presentational, formatting, and stylistic elements of a message Example: This coffee is amazing. ☕

Analytics with Photo & Video Data

Visual Sentiment

Text based sentiment analysis is not always enough to understand the sentiments of users.

Visual Data (Photos and Videos) can also be analyzed and classified based on their sentiment (positivity)



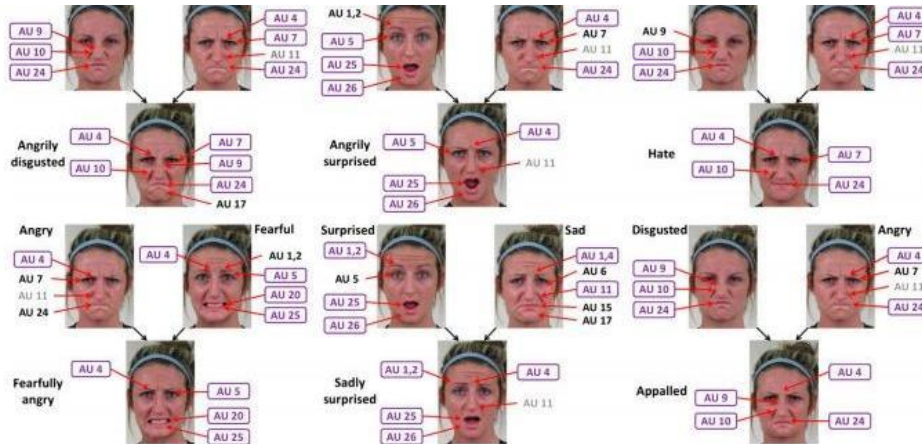
Analytics with Photo & Video Data

Visual Sentiment

Facial Expressions are relatively easier to detect through Machine Learning/Deep Learning techniques

Research focuses on six basic categories of emotions — **happiness, surprise, anger, sadness, fear, and disgust**.

Research (Du et al 2014) defines 21 emotion categories. Sample images of their facial expressions were collected from 230 human subjects.



Analytics with Photo & Video Data

Visual Sentiment

Challenges in Visual Sentiment Analysis:

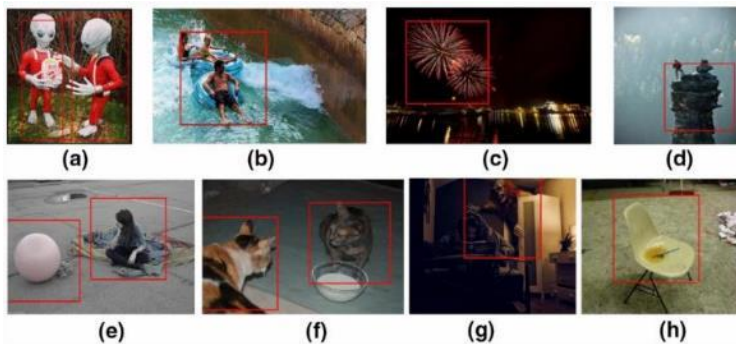
Visual semantic is **hidden in images** (not expressed with words – but facial expression <easy>, body language or ambience: to be detected)

Visual sentiment = Visual semantic, and there is **no high level visual semantic dictionary** like text-analysis lexicons

Visual sentiments from images, requires high level visual semantic ontology features instead of low-level visual features

Example (Food for Thought)

Many people take smiling photos at night. In this scene, dark night and simile are two different visual sentiments. In general, simile contains positive sentiment and dark night contains negative sentiment. As we can understand, simile relates to the main topic of this kind of images, while dark night is the background which is far from the main topic of the scene. BUT: if no smile – DARK can well be the main topic !

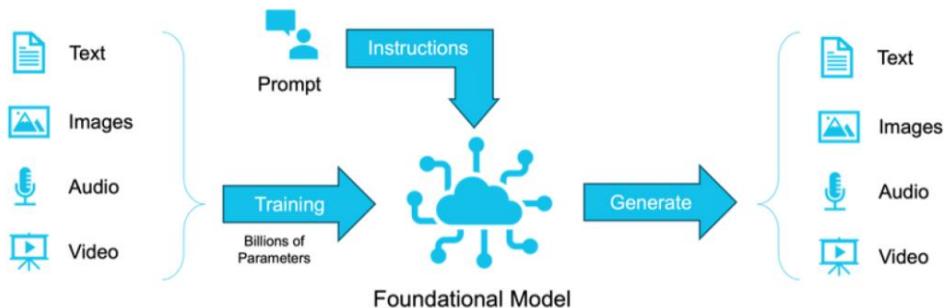


Generative AI in Marketing Analytics

Example: Chat GPT & Use in Marketing

GenAI: Key definitions

- **Generative AI (GenAI)** refers to artificial intelligence systems that can generate new content—such as texts, images, audio, and video—in response to prompts by a user, after being trained on an earlier set of data.



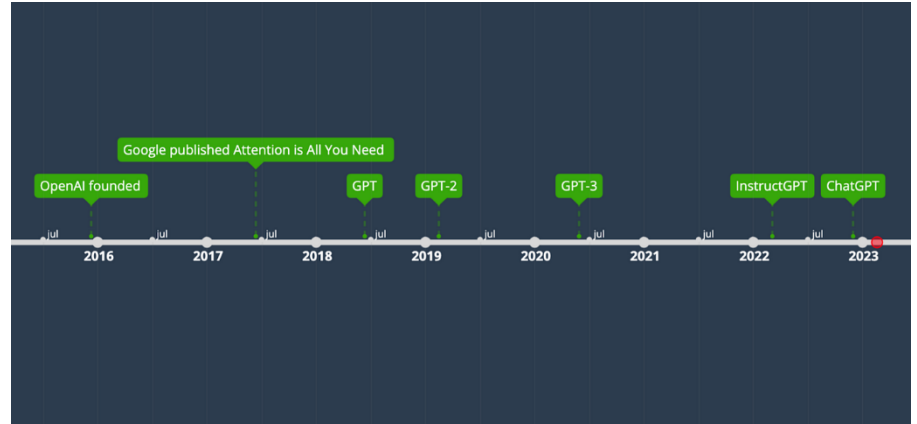
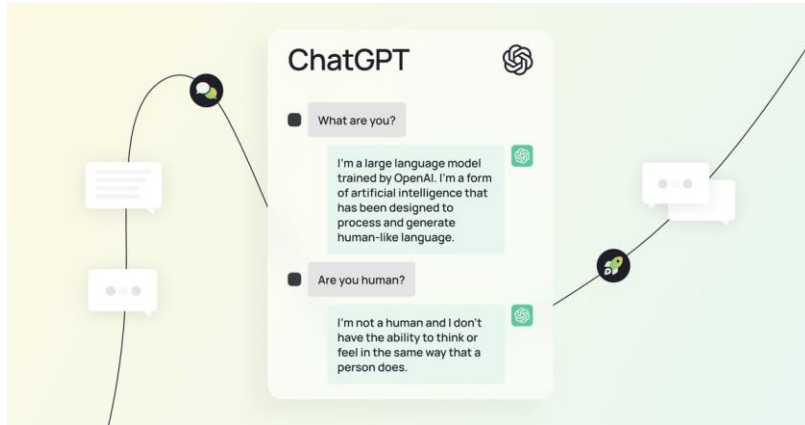
- **Training** is the *process* by which a machine learning model learns to perform a specific task.
- **Models** are the computational structure and algorithms that *enable Generative AI to process data, learn patterns, and perform tasks such as generating text, images, or making decisions.*
- **Parameters** are the *internal variables of an AI model that are learned from the training data.* These parameters are the core components that define the behaviour of the model and determine how it processes input data to produce output

ChatGPT: AI (NLP) Based Generative AI Tool by OpenAI

ChatGPT is an artificial intelligence (AI) chatbot developed by OpenAI and **released in November 2022**.

It is built on top of **OpenAI's GPT families of large language models (LLMs)** and has been fine-tuned (an approach to transfer learning) using both supervised and reinforcement learning techniques.

How many of you actively use GPT ? How many of you the paid version ?



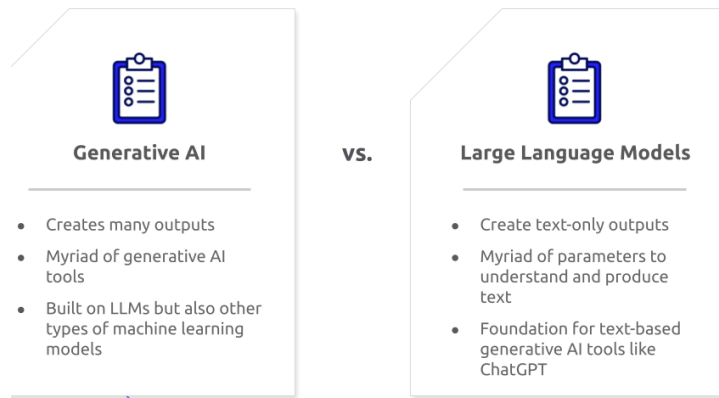
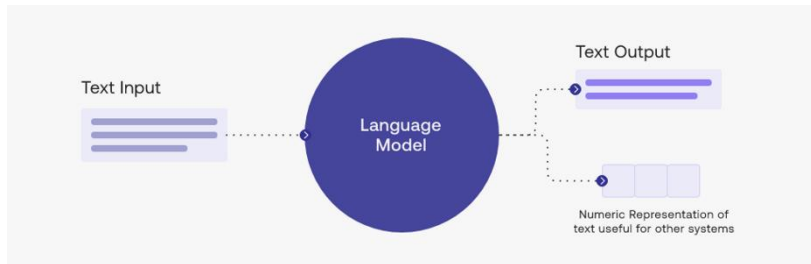
Gen-AI: ChatGPT & Large Language Models

LLMs (Large Language Models) are fed with large amounts of text data from across the internet wikipedia, scientific articles, books, blogs, forums, websites etc., to train so it can generate new content similar to the one it has been trained on.

These models can solve Natural Language use cases such as **question / answering, summarization, writing new content, generating code, and performing sentiment analysis**. Some examples: GPT, BERT.

ChatGPT is a large language model (LLM)

that uses natural language processing (NLP) and machine learning algorithms to generate human-like responses to user inputs.



Source Figure 1: <https://www.dallasnews.com/opinion/commentary/2023/02/23/editorial-cartoon-chatgpt/> Hariri, W. (2023). Unlocking the Potential of ChatGPT: arXiv preprint arXiv:2304.02017.

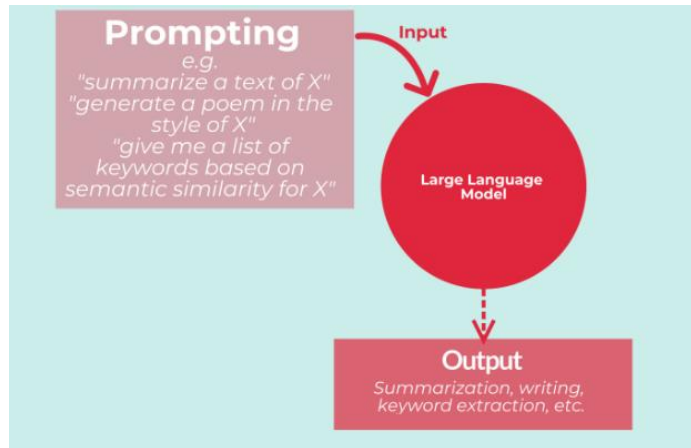
Source Figure 2: <https://www.adlittle.com/cn-en/insights/report/generative-artificial-intelligence-toward-new-civilization>
Vaswani, A., Shazeer, N. et al. (2017). Attention is all you need. *Advances in neural information processing systems*, 30.

ChatGPT: How Does it Work – or – Prompt Engineering

ChatGPT prompts are the textual inputs (e.g., questions, instructions) that you enter into ChatGPT to get responses.

A more specific and carefully worded prompt will get you better and more accurate responses.

Prompt Engineering is a new discipline for developing and optimizing prompts to efficiently use language models (LMs)

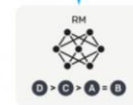


A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



A new prompt is sampled from the dataset.

The policy generates an output.



Once upon a time...

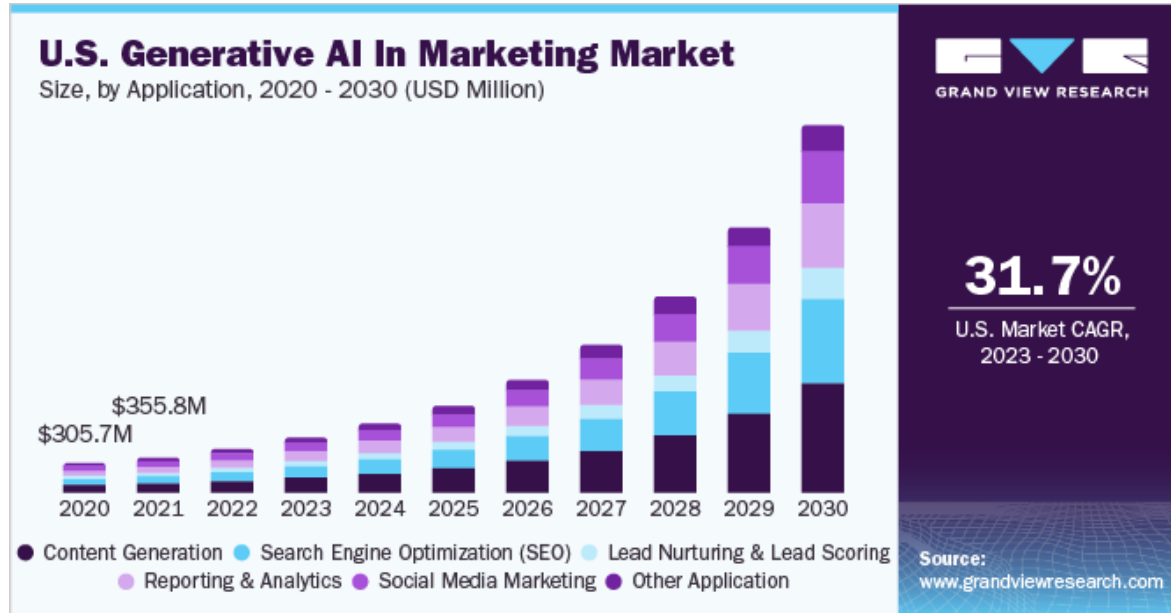
The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



r_k

GenAI: Growth & Role in Digital Marketing



ChatGPT: Why you should care about it?

Use of Generative AI in Marketing is rapidly increasing.

Examples: AI Generated content (text & images), AI generated product profiles and solutions > brings very interesting, yet unsolved research questions:

*The effect of **Chat GPT generated vs Human Generated short ad texts** on social media – in terms of likes, number of comments etc.*

*Which one do you think is more influential – of course: it **depends**. But depends on what – product, brand, audience, other factors ?*

AI Generated Ad Visuals



GPT Generated Ad Texts



Thank You